A/C ORIFICE TUBE (Continued)

and helps to reduce the potential for blockage of the metering orifice by refrigerant system contaminants (Fig. 14). The outlet end of the tube has a nylon mesh diffuser screen. The O-rings on the plastic body of the tube seal it to the inside of the liquid line and prevent the refrigerant from bypassing the fixed metering orifice.

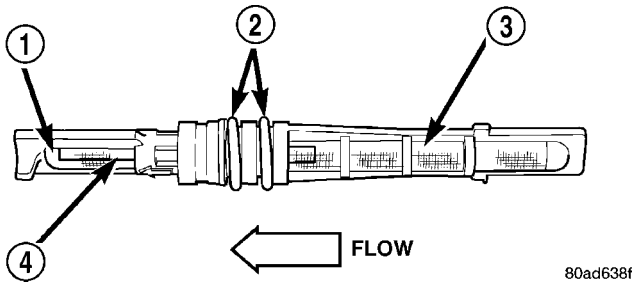

Fig. 14 Fixed A/C Orifice Tube - Typical

1 - DIFFUSER SCREEN
2 - O-RING SEALS
3 - INLET FILTER SCREEN
4 - ORIFICE

OPERATION

The fixed A/C orifice tube is used to meter the flow of liquid refrigerant into the A/C evaporator. The high-pressure liquid refrigerant from the A/C condenser expands into a low-pressure liquid as it passes through the metering orifice and diffuser screen of the A/C orifice tube.

The A/C orifice tube is not serviceable. It cannot be repaired, and if faulty or plugged, it must be replaced as part of the A/C liquid line (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/LIQUID LINE - REMOVAL).

DIAGNOSIS AND TESTING

A/C ORIFICE TUBE

WARNING: The A/C liquid line between the A/C condenser and the A/C orifice tube can become hot enough to burn the skin. Use extreme caution when performing the following test to prevent possible personal injury.

NOTE: The A/C orifice tube can be checked for proper operation using the following procedure. However, the A/C orifice tube is only serviced as a part of the A/C liquid line. If the results of this test indicate that the A/C orifice tube is obstructed or missing, the A/C liquid line must be replaced.

(1) Confirm that the refrigerant system is properly charged (Refer to 24 - HEATING & AIR CONDITIONING - DIAGNOSIS AND TESTING - A/C PERFORMANCE TEST).

(2) Start the engine. Turn on the A/C system and confirm that the compressor clutch is engaged.

(3) Allow the A/C system to operate for five minutes.

(4) Lightly and cautiously touch the A/C liquid line near the condenser outlet at the front of the engine compartment. The liquid line should be hot to the touch.

(5) Touch the A/C liquid line near the evaporator inlet at the rear of the engine compartment. The liquid line should be cold to the touch.

(6) If there is a distinct temperature differential between the two ends of the A/C liquid line, the A/C orifice tube is in good condition. If there is little or no detectable temperature differential between the two ends of the A/C liquid line, the A/C orifice tube is obstructed or missing and the A/C liquid line must be replaced (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING/A/C LIQUID LINE - REMOVAL).

ACCUMULATOR

DESCRIPTION

The A/C accumulator (Fig. 15) is mounted in the engine compartment between the A/C evaporator and the A/C compressor. An integral mounting bracket is used to secure the A/C accumulator to the frame rail. The A/C low pressure switch is mounted on a fitting located on the top of the A/C accumulator.

The A/C accumulator cannot be repaired and, if faulty or damaged, it must be replaced. The rubber O-rings used on accumulator connections and the A/C low pressure switch are available for service replacement.

OPERATION

Refrigerant enters the A/C accumulator mostly as a low pressure vapor through the inlet tube. Any liquid, oil-laden refrigerant falls to the bottom of the canister, which acts as a separator. A desiccant bag is mounted inside the accumulator canister to absorb any moisture which may have entered and become trapped within the refrigerant system A filter is also mounted inside the canister to trap any foreign material that may have entered the refrigerant system during assembly. The low pressure vapor exits the A/C accumulator through the outlet tube.

The A/C accumulator cannot be repaired. If the A/C accumulator is faulty or damaged, or if the refrigerant system has been contaminated or left open to the atmosphere for an indeterminable period or if the A/C compressor has failed, it must be replaced.